

Analysis of Health Variables in Smokers vs Never-Smokers

A comparison of triglyceride levels and height

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Introduction

Smoking is associated with many negative health outcomes, including cardiovascular disease. Public health datasets allow comparisons of physiological variables between smokers and never-smokers. The focus of this analysis is specifically on triglyceride levels, because elevated triglyceride concentration in blood are associated with cardiovascualar disease and strokes.

A public health-screening data set [1] was therefore analyzed to answer the question: **Do smokers and never-smokers differ in triglyceride levels and height?** Height was chosen only as a comparison variable and not expected to differ significantly between the groups. On the other hand, triglyceride levels were hypothesised to differ. Furthermore, male and female participants are considered separately to explore if differences in sex composition influence the results

Data sample

The dataset comes from the Korean National Health Insurance Service and was dowloaded from Kaggle [1]. It contains 991,346 health-screening records. The dataset variables relevant to the analysis were smoking status, height (cm), triglyceride levels (mg/dL) and sex (female or male). The analysis focused on comparing current-smokers (category 3) and never-smokers (category 1), while former-smokers (category 2) were excluded. The analysis was perfomed both for the the full selected sample as well as separately for male and female participants.

To compare current- and never-smokers, each group was divided into 50 subsamples. The mean of each group was calculated as the mean of the 50 subsample means, while its uncertainty was calculated using their standard deviation. To compare the two groups, the differences between corresponding subsample means were calculated. The mean of these 50 mean differences was calculated as the overall difference between the two groups and their standard deviation was used as its statistical uncertainty. The absolute mean diffence was then divided by its statistical uncertainty and results above 3 sigma were considered statistically significant.

Results and discussion

1. Height results

Initially, the overall difference between height of current-smokers and never-smokers was calculated as:

$$\Delta h = (9.79 \pm 0.13)\,\mathrm{cm} \tag{1}$$

Because this initial result was unexpected, a variable confounding the results was suspected. Therefore, the sex composition of current-smokers and never-smokers was examined. The results are shown in Fig. 1.

Since 92.4 % of current-smokers were male and 72.7 % of never-smokers were female, it was hypothesised that sex composition was counfouding the results. The analysis was therefore performed again separately for male and female participants. The height difference decreased to 1.19 cm for men and 1.87 cm for women, as shown in Fig. 2. This suggests that the large overall height difference was mainly influenced by the unequal sex composition of the two smoking groups, rather than smoking status alone.

The overall height difference was statistically significant (78.11 σ). The differences remained statistically significant when analysing men (7.99 σ) and women (6.26 σ) separately. Although the height differences remained statistically significant when analysing men and women separately, they were much smaller than the overall difference. These remaining differences could be explained by other confounding variables, such as age or socioeconomic background which were not considered in this analysis. That's why the results do not necessarily indicate that smoking itself affects height.

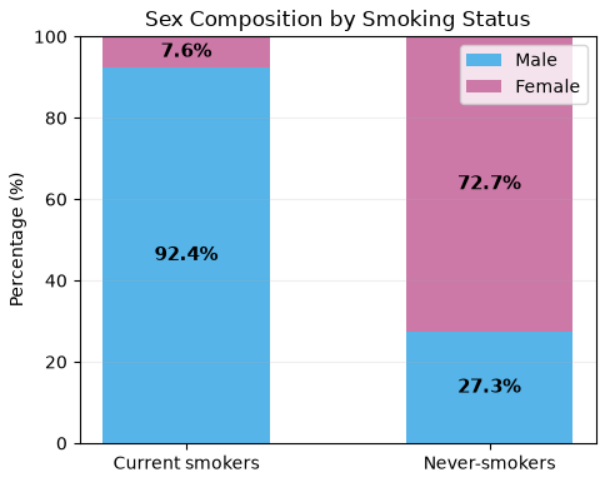


Figure 1: Sex composition of current-smokers and never-smokers.

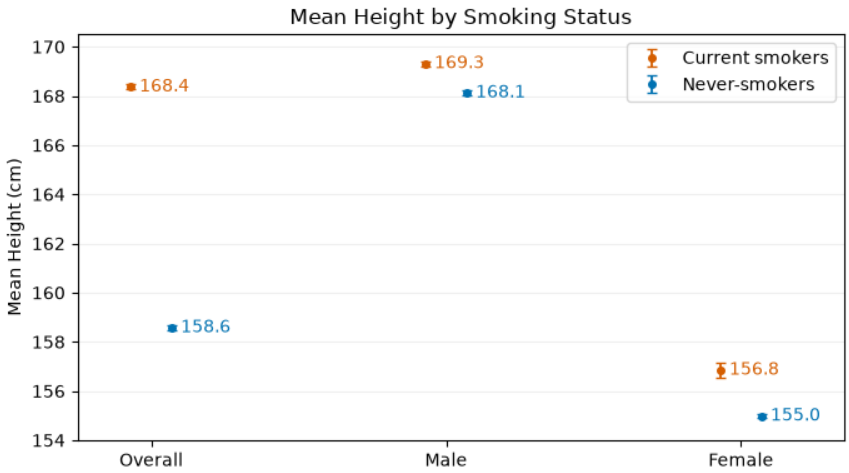


Figure 2: Mean height (cm) of current-smokers and never-smokers, shown for the overall sample and separately for male and female participants. Error bars represent the standard deviation of the 50 subsample means.

2. Triglyceride levels

Overall, current-smokers had higher mean triglyceride levels than never-smokers by 52.70 ± 1.83 mg/dL. This difference was statistically significant ($28.73\ \sigma$). After analysing men and women separately, the differences decreased but remained statistically significant: 37.11 ± 2.39 mg/dL for men ($15.52\ \sigma$) and 18.14 ± 5.48 mg/dL for women ($3.31\ \sigma$). All three comparisons exceeded 3 sigma. These results suggest an association between smoking status and higher triglyceride levels, even after accounting for sex composition. However, other factors, such as age and lifestyle, may also influence the observed differences.

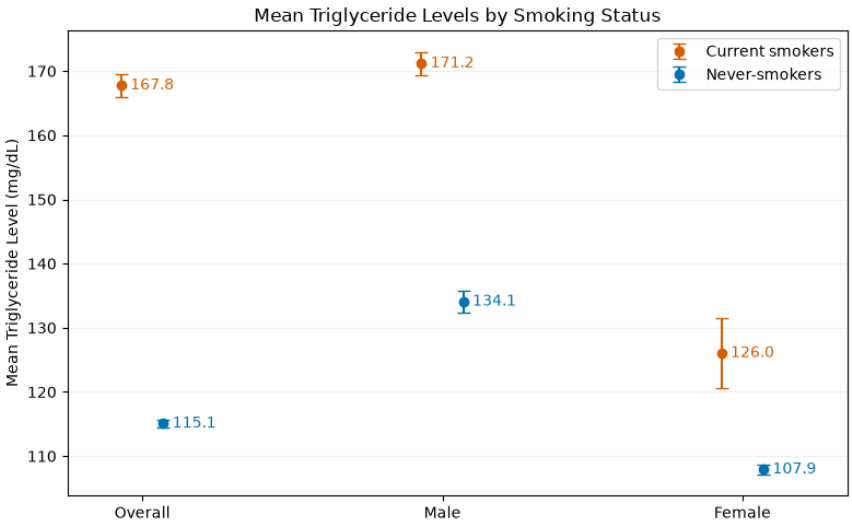


Figure 3: Mean triglyceride levels (mg/dL) for current-smokers and never-smokers, shown for the overall sample and separately for male and female participants. Error bars represent the standard deviation of the 50 subsample means.

To conclude, current-smokers showed higher triglyceride levels than never-smokers. The initial enormous height difference can be largely explained by the unequal sex composition of current-smokers and never-smokers. These results show that caution is needed when interpreting observational data and considering potential confounding variables.

References

[1] National Health Insurance Service in Korea. *Smoking and drinking data set*. Available at: https://www.kaggle.com/datasets/sooyoungher/smoking-drinking-dataset/data.